

Executive Hotel Booking EDA Report

Day 15 Data Science Assignment

End-to-end Exploratory Data Analysis, data-quality assessment, business interpretation, and management recommendations.

Final analytical dataset: 25,000 unique bookings

Primary business issue: 72.18% cancellation rate

1. Executive Summary

This report presents an end-to-end Exploratory Data Analysis of hotel booking records. The original dataset contained **25,180 records and 35 variables**. After removing 180 exact duplicate records and applying the documented cleaning and preprocessing steps, the final analytical dataset contains **25,000 unique bookings**.

The most important executive finding is the high overall cancellation rate of **72.18%**. Cancellation risk also rises materially with booking lead time, while Online TA represents a large-volume segment with relatively high cancellation. Resort Hotel has a higher average ADR than City Hotel, whereas City Hotel contributes greater booking volume.

Operationally, **11.95%** of bookings show a difference between reserved and assigned room type. This should be monitored to distinguish planned upgrades from availability or service issues.

Key Performance Indicators

KPI	Result
Unique bookings	25,000
Cancellation rate	72.18%
Average ADR	140.59
Average stay	3.69 nights
Average satisfaction score	3.85
Estimated revenue	13,384,199.45

2. Dataset and Data Quality Assessment

The dataset contains booking, hotel, date, customer, room, channel, cancellation, satisfaction, and revenue-related fields. Both numerical and categorical variables were assessed.

Data-quality findings

Issue	Finding	Treatment
Duplicate records	180 exact duplicates	Removed before analysis
Missing Company_ID	20,729	Filled with 'Unknown'
Missing Country	466	Filled with 'Unknown'
Missing Agent_ID	304	Filled with 'Unknown'
Missing Meal_Type	245	Filled with 'Unknown'
Missing Satisfaction_Score	164	Missing/invalid values handled; median imputation
Missing Children	121	Filled with 0
Missing Hotel_Location	106	Filled with 'Unknown'
Missing ADR	82	Median imputation
Invalid satisfaction value	1 value above 5	Converted to missing, then median-imputed
Date inconsistency	6 arrivals before booking date	Flagged for data-quality review
Total nights inconsistency	0 record	Total_Nights derived from weekend + weekday nights

Outlier treatment: IQR-based analysis was used to identify potential outliers. Extreme ADR, stay length, lead time, or revenue values were investigated rather than automatically deleted, because legitimate premium or extended bookings can appear as statistical outliers.

3. Cleaning and Analytical Methodology

The analytical workflow followed the EDA process used in the notebook: understand the data, assess quality, clean and preprocess, summarize distributions, compare groups, examine relationships, visualize patterns, and translate findings into business actions.

Key preprocessing steps

- Exact duplicate booking records were removed.
- Date fields were converted to datetime format.
- Missing categorical and identifier fields were handled using 'Unknown' where appropriate.
- Missing Children values were treated as zero.
- Invalid Satisfaction_Score values above the valid scale were converted to missing and median-imputed.
- Missing ADR values were median-imputed.
- Total_Nights was derived from Weekend_Nights + Weekday_Nights.
- Categorical inconsistencies and whitespace were standardized.
- Analytical features such as Lead_Time_Group, Cancellation_Status, Arrival_Year, Arrival_Month, and Room_Changed were created.

Analyses performed

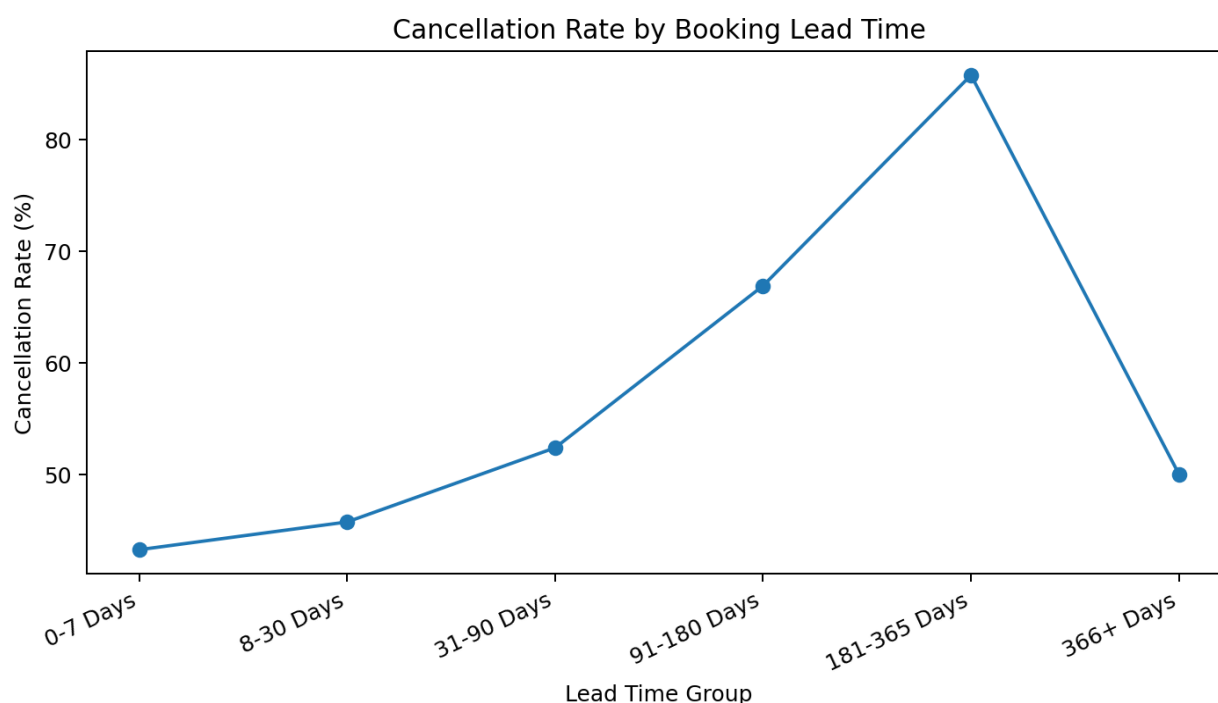
Descriptive statistics, univariate analysis, outlier analysis, bivariate analysis, group-wise analysis, correlation analysis, and Matplotlib/Seaborn visualizations were used to identify meaningful patterns.

4. Key EDA Findings

4.1 Cancellation and lead time

Overall cancellation is **72.18%**. The lead-time analysis shows a clear increasing pattern: short-lead bookings have lower cancellation, while bookings made many months in advance show much higher cancellation.

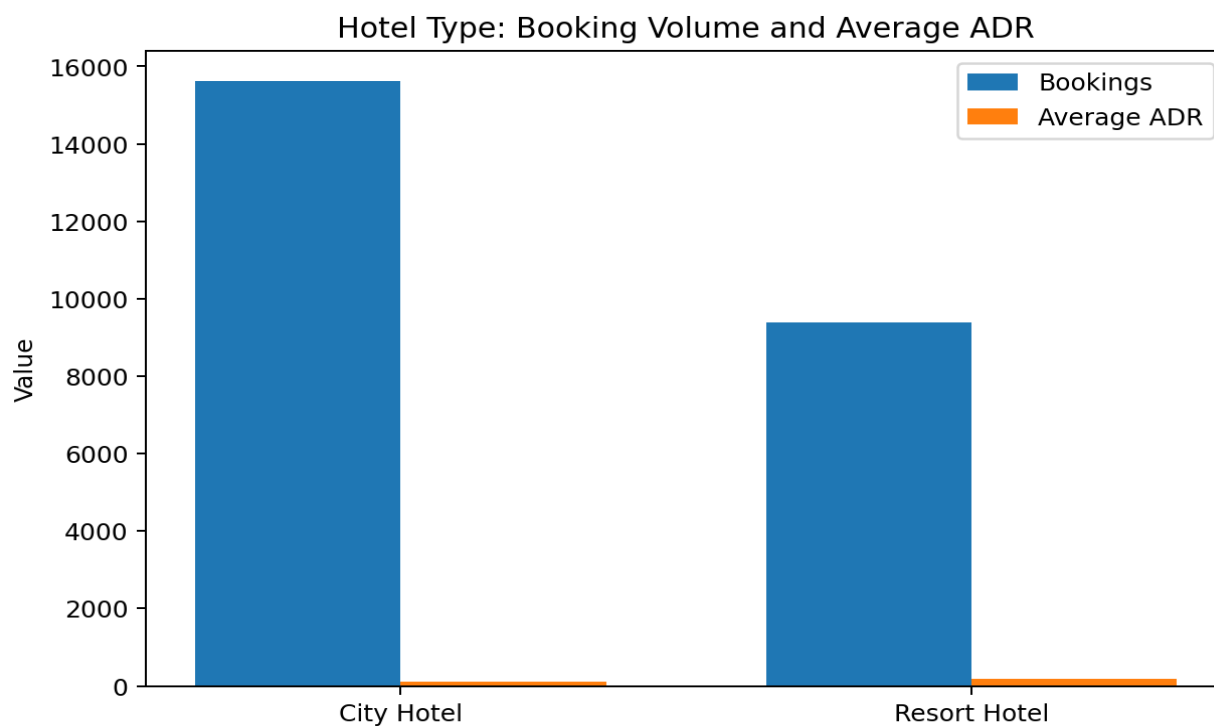
Lead time	Bookings	Cancellation rate
0-7 Days	511	43.2%
8-30 Days	1,594	45.7%
31-90 Days	4,122	52.4%
91-180 Days	6,171	66.9%
181-365 Days	12,600	85.8%
366+ Days	2	50.0%



4.2 Hotel type performance

City Hotel has the greater booking volume, while Resort Hotel has the higher average ADR. The different commercial profiles support hotel-specific revenue and demand-management strategies.

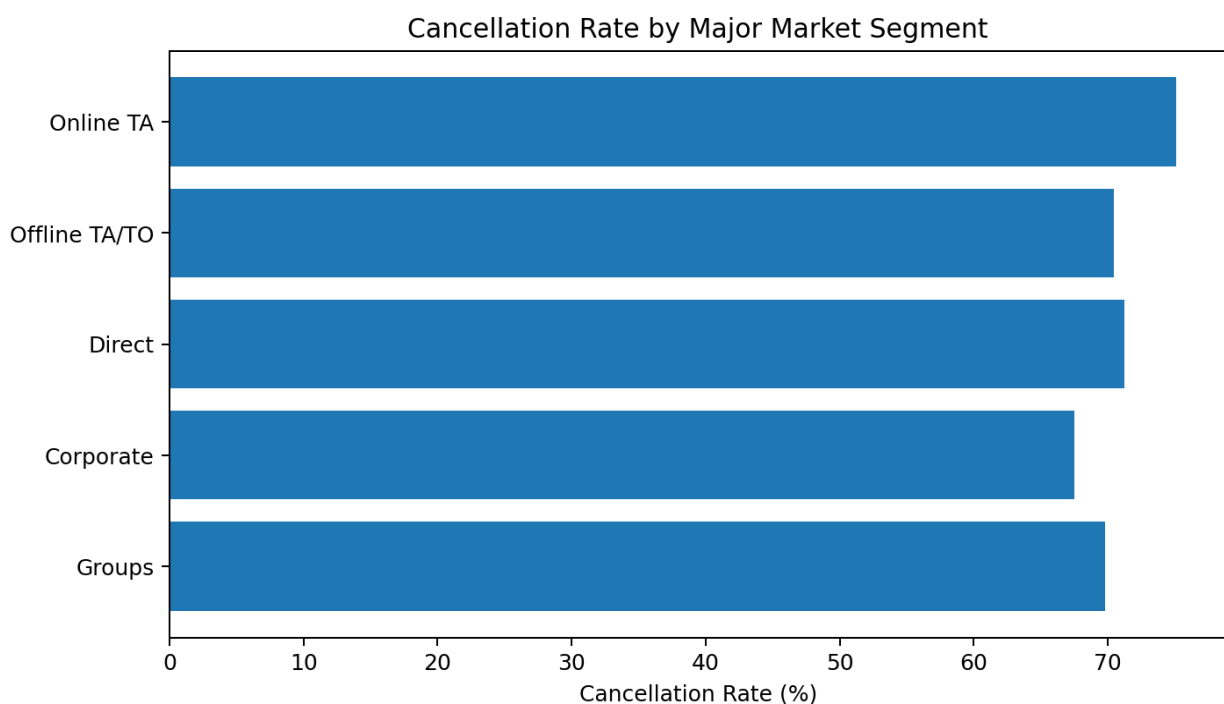
Hotel type	Bookings	Cancellation rate	Average ADR
City Hotel	15,611	73.3%	120.76
Resort Hotel	9,389	70.3%	173.56



4.3 Market segment and deposit pattern

Online TA is the largest market segment with approximately 10,852 bookings and a cancellation rate of 75.1%. Its combination of scale and cancellation risk makes it strategically important.

Market segment	Bookings	Cancellation rate
Online TA	10,852	75.1%
Offline TA/TO	5,440	70.4%
Direct	3,808	71.2%
Corporate	2,990	67.5%
Groups	1,910	69.8%



The Non Refund deposit category shows an unusually high cancellation rate of approximately 93.0%. This is an important anomaly for management investigation, but the EDA does not establish causation.

5. Key Business Insights

Insight 1 — High overall cancellation rate. The final dataset contains 25,000 unique bookings, with 72.18% canceled. This represents a major opportunity for improving booking stability and revenue predictability.

Insight 2 — Longer lead time is associated with higher cancellation. Cancellation rises from roughly 43% for 0–7 day lead times to roughly 86% for 181–365 day lead times. Long-lead bookings therefore warrant stronger confirmation and retention processes.

Insight 3 — Online TA is a high-volume, high-risk segment. Online TA contributes the largest booking volume and also shows a relatively high cancellation rate, making channel-specific management important.

Insight 4 — Resort Hotel has higher ADR while City Hotel has higher volume. This suggests that pricing, promotion, and demand-management strategies should differ by hotel type.

Insight 5 — Room assignment changes are operationally meaningful. Approximately 11.95% of bookings have a different assigned room from the reserved room. This should be monitored to distinguish planned upgrades from availability constraints or service issues.

6. Management Recommendations

- 1. Strengthen confirmation for long-lead bookings.** Use proactive confirmation and reconfirmation for bookings made several months before arrival.
- 2. Implement cancellation-risk management.** Prioritize bookings using lead time, hotel type, market segment, deposit type, and distribution channel.
- 3. Strengthen OTA booking management.** Review cancellation behavior by OTA and use channel-specific communication and retention practices.
- 4. Investigate the Non Refund cancellation anomaly.** Review deposit policies, payment workflows, cancellation coding, and booking processes before changing policy.
- 5. Apply hotel-specific pricing strategies.** Use Resort Hotel's higher ADR profile for value-based pricing and packages while focusing City Hotel strategy on volume and occupancy.
- 6. Monitor room assignment changes.** Track reserved-versus-assigned room differences and investigate recurring operational causes.
- 7. Establish regular KPI monitoring.** Track cancellation rate, ADR, stay length, satisfaction, channel performance, and revenue indicators on a recurring basis.

7. Conclusion

The analysis of 25,000 cleaned hotel booking records provides a clear view of booking behavior, cancellation risk, pricing, customer satisfaction, market segments, and operational characteristics.

The dominant business issue is the high cancellation rate of 72.18%, with cancellation increasing substantially as booking lead time becomes longer. Online TA is an important high-volume segment with elevated cancellation risk. Resort Hotel records a higher average ADR, while City Hotel contributes greater booking volume.

Operationally, approximately 11.95% of bookings show a reserved-versus-assigned room difference. Together, these findings support targeted cancellation management, channel strategy, hotel-specific pricing, operational monitoring, and regular KPI review.

Final KPI Snapshot

Metric	Final result
Dataset after duplicate removal	25,000 bookings
Cancellation rate	72.18%
Average ADR	140.59
Average stay	3.69 nights
Average satisfaction	3.85 / 5
Estimated revenue	13,384,199.45
Room assignment change rate	11.95%

Prepared from the completed Day 15 Colab EDA notebook and the provided hotel booking dataset.